

# Patrick Wagstrom

Data Science, Engineering, and Artificial Intelligence Leader  
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I am an innovative hands-on executive leader of data, machine learning, and software engineering organizations with broad experience in foundational research, finance, telecom, and video streaming. I build organizations that transform companies and markets with data driven insights and best of breed emerging technologies.

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## Professional Experience

### **Brightcove, Boston, MA and Hartford, CT**

#### **Chief Data Officer**

May 2021 - December 2022

- Executive responsible for all analytics, data, and machine learning within Brightcove including engineering, governance, policy, and vendor relations.
- Spearheaded end-to-end analytics collection and processing systems, handling trillions of rows of data and billions of monthly video views, utilizing AWS and GCP for cloud deployment.
- Managed an organization of 34 FTEs and 12 contractors, overseeing data platforms, analytics and insights, data and model governance, and customer-facing data products.
- Executed the \$12.3mm acquisition of Wicket Labs, expanding Brightcove's media analytics capabilities and growing the product customer base by over 50x in under a year.
- Collaborated closely with Brightcove and customer C-level executives, driving corporate strategies and fostering strong relationships.
- Championed initiatives to enhance engineering performance and implement effective performance management across the organization.
- Negotiated and owned a multi-million dollar partnership with Google around Google Cloud Platform.
- Maintained and built on multi-cloud data infrastructure using tools such as Trino, Glue, Lambda, SageMaker, and Redshift on AWS and BigQuery, Cloud Composer, Dataflow, and Pub/Sub on GCP.
- Developed multi-modal video classification models using text transcripts, audio, and video processing.
- Directly conducted research on the impact of super-resolution algorithms on streaming, from holistic bandwidth, encoding cost, and device playback cost perspectives.

### **Verizon, Basking Ridge, NJ and Hartford, CT**

#### **Director of Emerging Technology**

September 2019-April 2021

- Built and led a team of 15 FTEs and 51 vendor contractors with a \$15mm annual budget, collaborating closely with CIO, CTO, and CDAO to influence corporate strategies.
- Managed a successful migration of data architecture from ad-hoc on-prem and Hadoop solutions to a streaming data architecture with Google BigQuery, demonstrating adept stakeholder management.
- Defined and architected an enterprise strategy for reproducible machine learning and MLOps, encompassing data management, model training, model serving, model refit, and model risk management.
- Developed cell site energy efficiency machine learning models, resulting in \$5mm annual savings and the creation of a digital twin for future simulation.
- Led the development of a holistic model connecting online brand discussions to customer profiles, supporting customer service operations and building patent-pending bot detection models.
- Worked alongside legal and policy experts to create several Verizon policies, including facial recognition, energy efficiency, model risk management, and data privacy.

- Developed systems for blockchain based identity management and sim swap protection across mobile carriers.

## **Capital One, New York, NY and McLean, VA**

### **Senior Director of Data Science for Machine Learning Platforms**

January 2019-September 2019

### **Director of Data Science for Machine Intelligence**

November 2016-January 2019

- Architect and overall lead of 37 people building the Capital One Card Machine Learning Platform - a platform for scalable and personalized real-time reinforcement learning based machine learning models on top of Kubernetes, gRPC, and AWS in a heavily regulated environment.
- Led a team that built and deployed multi-modal customer intelligence models working on both structured and unstructured (text, voice) data resulting in \$27mm annual savings.
- Hired, managed, and developed a distributed team of 18 data scientists and data analysts.
- Built out a \$1.2mm academic research partnership with universities to explore implications of fairness in artificial intelligence.
- Conducted more than 300 hiring interviews of data scientists, data engineers, and product managers to help grow Card Machine Learning from 33 to 168 people and Capital One's New York Card team from 8 to nearly 200.
- Championed reproducible machine learning and MLOps across the organization, ensuring best practices in data management, model training, model serving, model refit, and model risk management.
- Collaborated with cross-functional teams to develop and implement the company's data privacy and AI ethics policies.

## **IBM Watson, Littleton, MA**

### **Research Staff Member/Technical Lead**

January 2015-November 2016

- Led a globally distributed team that built the IBM Watson Conversation service - an innovative service to create rich conversational interactions using natural language processing, entity recognition, and scripted dialog.
- Global team lead for Watson Developer Cloud Tooling. Created cutting edge applications for creating, training, and maintaining cognitive and machine learning solutions including Watson Engagement Advisor and IBM Watson Natural Language Classifier.
- Conducted research on optimal methods to build and configure cognitive and machine learning systems resulting in a 95% reduction in human time required to train these systems.
- As a subject matter expert and research scientist, traveled around the world to conduct research with and work side-by-side with our customers.
- Led the charge to evolve internal development standards from a legacy Java stack to a modern stack based on Angular, Node.js, PostgreSQL, and more.

## **IBM Watson - Watson Life, Yorktown Heights, NY**

### **Research Staff Member**

January 2014-January 2015

- Founding member of the IBM Watson business group within IBM
- Leadership team member for an innovative new lab within IBM responsible for evaluating promising consumer applications of cognitive computing.
- Organized and led globally distributed teams on how to contextualize solutions to cognitive computing challenges.
- Engineering and on-side lead for the IBM Food Truck at SXSW, which demonstrated generative AI based cognitive computing to more than 4,000 people and resulted in more than 1 billion media impressions.

## **IBM TJ Watson Research Center, Yorktown Heights, NY**

### **Research Staff Member**

August 2009-January 2014

- Team lead for the "Millennial Enterprise" aspect of IBM's 2013 Global Technology Outlook, which described the need for user data in building out AI/ML and led to me being a founding research scientist for IBM Watson Group.
- Analytics lead for JazzHub, IBM's cloud software development strategy. Designed analytics strategy, implemented metrics driven development, introduced A/B testing, and developed analytics dashboards.
- Strategic and technical contributor to IBM's acquisition process for \$N different companies.
- Developed and designed GitMiner - an open source project used by 15 universities to perform graph analysis on large scale software engineering databases such as GitHub and BitBucket.
- Published papers on topics around distributed collaboration, technical debt in software, and flow of ideas in software engineering communities.
- Mentored three incredible Ph.D. student interns on projects related to collaboration in software engineering

## **Carnegie Mellon University, Pittsburgh, PA**

### **Graduate Research Assistant**

August 2003-July 2009

- Conducted research on the interactions between people and firms in the development of open source software in a dual Ph.D. degree program between the School of Computer Science and College of Engineering.
- Designed and developed CVSMiner an open source tool to perform social network and technical analysis of software engineering ecosystems such as GNOME and Eclipse.
- Worked with members of the GNOME Foundation and Eclipse Foundation to evaluate and improve the relationships between non-profit foundations that manage open source ecosystems and commercial firms.
- Utilized a variety of qualitative and quantitative research methods: stakeholder interviews, message analysis, natural language processing, data mining, machine learning, and social network analysis to generate insight into largely ad hoc software development processes.

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## **Education**

### **Ph.D. in Engineering and Public Policy and Computation, Organizations, and Society, May 2009**

Carnegie Mellon University, Pittsburgh, PA

Thesis: "Vertical Interaction in Open Software Engineering Communities". Advisors: Dr. James Herbsleb and Dr. Kathleen Carley.

### **MS in Computation, Organizations, and Society, May 2007**

Carnegie Mellon University, Pittsburgh, PA

### **MS in Computer Science, August 2003**

Illinois Institute of Technology, Chicago, IL

Thesis: "Scarlet: A Framework for Context Aware Computing". Advisor: Dr. Xian-He Sun.

### **BS in Computer Science / BS in Computer Engineering / BS in Electrical Engineering, May 2002**

Illinois Institute of Technology, Chicago, IL